SABA KHARABADZE, PHD

sabakharabadze.com | s.kharabadze@gmail.com | linkedin.com/in/kharabadze | Google Scholar

Pisa, Tuscany, Italy

SUMMARY

ML Research Engineer with a PhD in Physics and 7+ years building scalable ML-enabled scientific and production systems. Expert in distributed workflows (JAX/PyTorch), Transformer architectures, and large-scale data processing (1TB+). Track record includes developing high-throughput ML discovery pipelines and designing an award-winning Transformer architecture for complex signal processing.

SKILLS

Programming: Python (PyTorch, JAX, NumPy, SciPy, pandas), C/C++, SQL, Bash, Git

Systems/Infra: Docker, Linux, HPC (GPU nodes, schedulers), CUDA, multi-GPU training, distributed training

Scientific/Domain: DFT (VASP, Gaussian), materials modeling, numerical simulation, signal processing

Languages: English (bilingual), Georgian (bilingual), Russian (proficient), German (basic), Italian (beginner)

RESEARCH ENGINEERING EXPERIENCE

ML Research Engineer (Research Scientist) - *CNIT RaSS Lab* - *Pisa, Italy* | April 2025 – Present

- Winner of the IEEE Radar Challenge: Developed a Transformer-based pipeline for heart rate estimation, achieving a 65% improvement over the official baseline.
- Build ML-based radar signal processing workflows for target detection and develop simulation frameworks for coverage analysis of mono- and multistatic radar systems.
- Implement reproducible experimentation code and analysis pipelines for model evaluation and system-level studies.

Physics AI Model Validation Expert - *Handshake MOVE Program* - *Remote* | August 2025 – Present

- Design complex physics reasoning problems and reference solutions to stress-test state-of-the-art AI model capabilities.
- Evaluate model responses and write targeted feedback to improve reasoning, logic, and physical correctness.

PhD Candidate - *Binghamton University (SUNY)* - *Binghamton, NY* | August 2018 – August 2024

- Built end-to-end pipelines for ML-accelerated atomistic simulation: data generation, model training, and high-throughput search.
- Scaled evolutionary search + ML potential workflows to 1M+ structures and 1TB energy/force data using HPC clusters.
- Developed Python tooling to transform raw simulation outputs into physical/electrochemical/structural properties used for discovery decisions.
- Implemented and validated an NPT barostat module in C for in-house molecular dynamics software (MAISE); achieved thermal expansion coefficients within 10% of experiment.
- Administered two university HPC clusters (1000+ CPU cores plus GPU nodes): installed/maintained compilers, CUDA, MPI, PyTorch stack, and scientific codes; enabled multi-node and multi-GPU use.

SELECTED PROJECT HIGHLIGHTS

- **ML-accelerated discovery of Li-Sn phases:** created DFT reference data (VASP), trained/used a neural network potential (C), and executed high-throughput evolutionary searches; identified 8 new stable structures and validated stability with phonons. Published in *npj Computational Materials* (a Nature Partner Journal).
- **Thermodynamic stability of Li-B-C compounds:** computed chemical potentials and performed quasi-harmonic phonon analysis to study stability across volumes and temperatures; published results in PCCP.
- **Engineering for reproducibility:** built scripts, Python tooling and deployed documentation website (Sphinx) for simulation workflows; maintained a public documentation site for MAISE.

EDUCATION

PhD in Physics, Binghamton University, State University of New York | May 2024
Dissertation: Machine Learning and *ab initio* insights into the design of lithium-based materials.

B.S. in Physics, Free University of Tbilisi | May 2017
Minor: Computer Science and Mathematics

SELECTED PUBLICATIONS

- Kharabadze, S. *et al.* (2022) Prediction of stable Li-Sn compounds: boosting ab initio searches with neural network potentials. *npj Comput Mater.*
- Kharabadze, S. *et al.* (2023) Thermodynamic stability of Li-B-C compounds from first principles. *Phys. Chem. Chem. Phys.*
- Hajinazar, S. *et al.* (2021) MAISE: Construction of neural network interatomic models and evolutionary structure optimization. *Comput. Phys. Commun.*

ADDITIONAL INDUSTRY EXPERIENCE

Founding Member, Data Science / ML - *Keepers* - https://kprs.pro - *Remote* | Dec 2024 – Present

- Build ETL pipelines to ingest exchange data into internal databases using Python, SQLAlchemy, and SQL; implement data validation and cleaning in pandas.
- Develop anomaly detection and trend prediction on financial time-series using XGBoost and feature engineering pipelines.
- Produce research analyses and monitoring reports to support strategy development and ongoing performance review.

Back-end Developer - *Eleven Wireless* - *Remote (Portland, OR)* | Oct 2017 – Apr 2018

- Built a Python middle layer interfacing SQL databases with ElasticSearch; deployed and maintained reporting service on AWS Linux.

Junior Business Analyst - *TBC Bank* - *Tbilisi, Georgia* | May 2017 – Feb 2018

- Produced system documentation and process diagrams within a Scrum-based innovations team.

HONORS

- Winner, IEEE Radar Challenge (International Competition) | 2025
- Handshake AI Fellow, *Handshake MOVE Program* | Aug 2025 – Present
- Travel grant to present at APS March Meeting, *Binghamton University* | 2022, 2023
- Bronze Medal, International Physics Olympiad (Tallinn, Estonia) | 2012